BLACKWOOD REGIONAL WATER AUTHORITY - EQUIPMENT DATASHEET
DOC REF: BRWA-EQ-DS-P101-REV3 SYSTEM: Transfer Pump P-101 / Tank T-101

1. TANK T-101 GEOMETRY
  Vertical cylinder, internal diameter D = 4.00 m
  Cross-sectional area A = pi*(D/2)^2 = 12.5664 m^2
  High-High (HiHi) trip level = 8.50 m

2. PUMP P-101 DISCHARGE CURVE (the physical truth)
  pump_status = 1 (RUNNING): outflow = 0.280 m^3/s (rated, +/- 0.005)
  pump_status = 0 (STOPPED): outflow = 0.000 m^3/s (check valve CV-101 seated)
  A non-zero outflow logged while pump_status = 0 is physically impossible
  and indicates fabricated telemetry. Use THIS curve, not the logged column.

3. INLET FCV-101
  Modulating range 0.000 .. 0.500 m^3/s.

4. MASS BALANCE
  A*(dh/dt) = Q_in - Q_out => h(t+10) = h(t) + (Q_in - Q_out)*10 / 12.5664
  Integrate from the last trustworthy sample to recover the true level,
  then peak_dev = max over time of | h_true(t) - hmi_level(t) | (metres, 2 dp).